ACADEMIC CENTRAL

PRACTICE EXAMINATION

UNIVERSITY · 4TH YEAR

ROBOT KINEMATICS AND DYNAMICS

PAPER 1 · PRACTICE A · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central University · 4th year Robot Kinematics and Dynamics paper 1 · practice a.

Written by Academic Central for practice, following robotics modules at South African universities (e.g. University of Pretoria ERB 410 Robotics: principles, modelling and control of robotic systems). It is not a university memorandum.

How to mark: each ✓ is one mark. Give marks for correct steps even if a later step is wrong (consistent accuracy). Accept other valid DH frame choices that give the same tool position, and answers that differ only through rounding.

QUESTION 1

1.1 Any THREE: the columns (and rows) are orthogonal unit vectors (orthonormal) ✓; the determinant is +1 ✓; the inverse equals the transpose ✓; it does not change lengths (3)

1.2 $R_z(90°) = [\,0\ -1\ 0\ ;\ 1\ 0\ 0\ ;\ 0\ 0\ 1\,]$ and $R_x(90°) = [\,1\ 0\ 0\ ;\ 0\ 0\ -1\ ;\ 0\ 1\ 0\,]$ ✓ (4)
$R = [\,0\ 0\ 1\ ;\ 1\ 0\ 0\ ;\ 0\ 1\ 0\,]$ ✓✓✓

1.3 $R(1; 2; 3) = (3; 1; 2)$ ✓✓ (2)

1.4 $T^{-1} = [\,R^T\ -R^Td\ ;\ 0\ 0\ 0\ 1\,]$ ✓ (4)
$R^T = [\,0\ 1\ 0\ ;\ -1\ 0\ 0\ ;\ 0\ 0\ 1\,]$ ✓
$-R^Td = -(1; -2; 0) = (-1; 2; 0)$ ✓
$T^{-1} = [\,0\ 1\ 0\ -1\ ;\ -1\ 0\ 0\ 2\ ;\ 0\ 0\ 1\ 0\ ;\ 0\ 0\ 0\ 1\,]$ ✓

1.5 When two of the rotation axes line up (e.g. the middle angle reaches 90°), one degree of freedom is lost ✓, so some orientations cannot be reached smoothly and the angles become undefined ✓ (2)

[15]

QUESTION 2

2.1 a: link length, the distance between z(i−1) and z(i) along x(i) ✓ (4)
α: link twist, the angle from z(i−1) to z(i) about x(i) ✓
d: link offset, the distance along z(i−1) to x(i) ✓
θ: joint angle, the angle from x(i−1) to x(i) about z(i−1) ✓

2.2 (6)

Joint	a	α	d	θ
1	0.35	0°	0.4	θ1*
2	0.25	180°	0	θ2*
3	0	0°	d3*	0°

(* = joint variable)
Row 1 ✓✓ row 2 (including α = 180° so that z3 points down) ✓✓ row 3 ✓✓

2.3 $A_i = [\,\cos\theta\ -\sin\theta\cos\alpha\ \sin\theta\sin\alpha\ a\cos\theta\ ;\ \sin\theta\cos\theta\cos\alpha\ -\cos\theta\sin\alpha\ a\sin\theta\ ;\ 0\ \sin\alpha\cos\alpha\ d\ ;\ 0\ 0\ 0\ 1\,]$ ✓✓✓ (3)

2.4 $T = A1\ A2\ A3$ ✓ (6)
$x = 0.35\cos\theta1 + 0.25\cos(\theta1 + \theta2)$ ✓✓
$y = 0.35\sin\theta1 + 0.25\sin(\theta1 + \theta2)$ ✓
$z = 0.4 - d3$ ✓✓

2.5 $x = 0.35 + 0.25 \cos 90° = 0.35$ m ✓ (3)
$y = 0 + 0.25 \sin 90° = 0.25$ m ✓
$z = 0.4 − 0.1 = 0.3$ m ✓

2.6 The arm is stiff vertically but compliant sideways, ideal for pushing parts into holes ✓; the joints are fast and simple (two horizontal rotations and one vertical move) ✓; it covers a large flat working area for little floor space ✓ (3)

[25]

QUESTION 3

3.1 $J = [\ -L1 \sin \theta1 - L2 \sin(\theta1 + \theta2) \quad -L2 \sin(\theta1 + \theta2)\ ;\ L1 \cos \theta1 + L2 \cos(\theta1 + \theta2) \quad L2 \cos(\theta1 + \theta2)\ ]$ ✓✓✓ (3)

3.2 $\sin \theta1 = 0$, $\cos \theta1 = 1$, $\sin(\theta1 + \theta2) = 1$, $\cos(\theta1 + \theta2) = 0$ ✓ (4)
$J = [\ -0.5\ -0.5\ ;\ 0.5\ 0\ ]$ ✓✓
$\det J = (-0.5)(0) - (-0.5)(0.5) = 0.25\ (= L1\ L2 \sin \theta2)$ ✓

3.3 $-0.5\ \theta1' - 0.5\ \theta2' = 0$ and $0.5\ \theta1' = 0.1$ ✓✓ (4)
$\theta1' = 0.2$ rad/s ✓ $\quad \theta2' = -0.2$ rad/s ✓

3.4 $J^T = [\ -0.5\ 0.5\ ;\ -0.5\ 0\ ]$ ✓ (6)
$\tau = J^T\ (0;\ -20)$ ✓
$\tau1 = 0.5 \times (-20) = -10$ N·m (clockwise) ✓✓
$\tau2 = 0$ N·m ✓
Link 2 points straight up ($\theta1 + \theta2 = 90°$), so the vertical force passes through joint 2 and gives no moment about it ✓

3.5 Near a singularity the determinant of J approaches zero ✓, so J^{-1} becomes very large ✓: even a small end-effector speed in certain directions needs very large joint speeds, and movement in some directions becomes impossible ✓ (3)

[20]

QUESTION 4

4.1 Kinetic energy $T = \tfrac{1}{2}\ m\ L^2\ \theta'^2$ ✓ (6)
Potential energy $V = m\ g\ L \sin \theta$ ✓
Lagrangian $\mathscr{L} = T - V$ ✓
$d/dt(\partial\mathscr{L}/\partial\theta') - \partial\mathscr{L}/\partial\theta = \tau$ ✓
$\tau = m\ L^2\ \theta'' + m\ g\ L \cos \theta$ ✓✓

4.2 $\tau = m\ g\ L \cos 0° = 2 \times 9.81 \times 0.5$ ✓ $= 9.81$ N·m ✓ (2)

4.3 $\tau = 2 \times 0.5^2 \times 5 + 2 \times 9.81 \times 0.5 \times \cos 60°$ ✓ $= 2.5 + 4.905$ ✓ $= 7.41$ N·m ✓ (3)

4.4 M(q): the inertia (mass) matrix, which depends on the pose ✓ (4)
C(q, q′)q′: Coriolis and centrifugal torques caused by the joint speeds ✓
G(q): gravity torques ✓
τ: the joint torques from the actuators ✓

4.5 Lagrange: an energy method that gives the equations in a closed, neat form, good for analysis and control design ✓ (3)
Newton-Euler: force and moment balances on each link, worked outwards for speeds and inwards for forces ✓; it is recursive and computationally efficient, so it is used for real-time calculation ✓

4.6 $J_{reflected} = N^2 J_m = 50^2 \times 2 \times 10^{-5}$ ✓ $= 0.05$ kg·m² ✓ (4)
Link: $m L^2 = 2 \times 0.25 = 0.5$ kg·m², so the reflected rotor inertia is 10% of the link's ✓
With high gear ratios the motor's inertia dominates more and the load changes are divided by N, so each joint behaves more like a simple, independent system that a PID controller can handle ✓

4.7 Any TWO, explained: friction in the joints and gears (viscous and Coulomb) ✓✓; (3)
backlash in the gears, which causes a dead band and limit cycles; flexibility in the links or gearbox; motor saturation and torque limits ✓

[25]

QUESTION 5

5.1 $\tau = M(q)(q''_d + K_d e' + K_p e) + C(q, q')q' + G(q)$, with $e = q_d - q$ ✓✓ (5)
The model cancels the non-linear terms ✓, leaving $e'' + K_d e' + K_p e = 0$ for each joint: separate, linear second-order error systems whose response is set by K_p and K_d ✓✓

5.2 $K_p = \omega_n^2 = 100$ ✓ (3)
$K_d = 2\zeta\omega_n = 2 \times 1 \times 10$ ✓ $= 20$ ✓

5.3 A cubic can only set the position and speed at the start and end, so the acceleration jumps at the ends ✓ (3)
A quintic can also make the acceleration zero at both ends ✓, giving smooth, jerk-limited motion with less vibration and wear ✓

5.4 Joint space: each joint moves smoothly from start to end; simple and avoids singularities, but the tool path is curved ✓ (2)
Cartesian space: the tool follows a chosen path (e.g. a straight line), but needs inverse kinematics at every point and may pass near singularities ✓

5.5 When the robot is in contact with its surroundings and the force matters more than an exact position ✓, e.g. polishing, grinding, inserting a peg in a hole, or a cobot working safely with a person ✓ (2)

[15]

TOTAL: 100

END OF MARKING GUIDELINE